

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Edu Genie application and use AI-powered learning features.	High
2	Authorization levels	All users (students) can access question answering, concept explanation, quiz generation, summarization, and learning recommendation features.	Medium
3	External interfaces	The application integrates with the Groq API (Llama 3.1 model) to generate AI-based educational responses.	High

4	Transactions processing	Transactions Processing	The system accepts the student's question, processes it through the Groq AI API, and returns an AI-generated response in real time.	High
5	Reporting	Reporting	The application displays generated answers, concept explanations, quizzes, summaries, and learning recommendations on the user interface.	Medium
6	Business rules, etc.	Business Rules	The system validates user input, processes only educational queries, and provides accurate, structured, and student-friendly responses based on the selected learning task.	High
7	Compliance to laws or regulations	Compliance to Laws or Regulations	The application does not store sensitive personal information and follows basic data privacy and secure API usage practices.	Medium
8	<i>Other</i>	Other	The application should provide a simple, responsive, and easy-to-use interface that supports beginners, intermediate, and advanced learners.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate AI responses quickly with minimal delay.	Response generated within 3–5 seconds under normal network conditions.

2	Scalability	The system should support multiple users without significant performance degradation.	Supports at least 100 concurrent users.
3	Security & Data Privacy	User queries and API communication should be secure, and sensitive information should not be stored.	Secure API communication (HTTPS) and no storage of sensitive user data.
4	Reliability & Availability	The application should remain available and function correctly with minimal downtime.	System availability of 99% during normal operation.
5	Usability & Accessibility	The application should provide a simple, responsive, and user-friendly interface for all learners.	Easy-to-use interface with responsive design compatible with desktop and mobile devices.
6	<i>Other</i>	The application should be easy to maintain, update, and deploy on different systems.	Compatible with Windows, Linux, and modern web browsers.